Complex From Trig

Michael Luo

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this explains so much about angie

aster after finding out that math prize for girls is 90 percent trig

1 Theory

Complex numbers are vectors.

Example 1.1 (HMMT Algebra 2018/6)**.** If

$$\cos\alpha + \cos\beta + \cos\gamma = \sin\alpha + \sin\beta + \sin\gamma = 1,$$

compute the smallest possible value of $\cos\alpha$.

Example 1.2 (2025 AIME I/8)**.** Let k be a real number such that the system

$$\begin{aligned} |25 + 20i - z| &= 5 \\ |z - 4 - k| &= |z - 3i - k| \end{aligned}$$

has exactly one complex solution z. Compute the sum of all possible values of k.

The great thing about complex numbers is that multiplication is angle addition, so they're better than normal vectors.

Example 1.3 (Machin's formula)**.** $\frac{\pi}{4} = 4\arctan\frac{1}{5} - \arctan\frac{1}{239}$.

Example 1.4. Compute

$$\sin 1^\circ + \sin 2^\circ + \cdots + \sin 179^\circ.$$

Example 1.5 (CMIMC Algebra 2018/7)**.** Compute

$$\sum_{k=0}^{2017} \frac{5 + \cos(\frac{\pi k}{1009})}{26 + 10\cos(\frac{\pi k}{1009})}.$$

2 Extras

If we have extra time, we can do this stuff I guess.

Lemma 2.1 (EGMO 6.28)**.** Let ABC be a triangle. It is equilateral if and only if $a^2 + b^2 + c^2 = ab + bc + ca$.

Example 2.2 (CMIMC Algebra 2016/6)**.** For some complex number ω with $|\omega| = 2016$, there is some real $\lambda > 1$ such that ω, ω^2, and $\lambda\omega$ form an equilateral triangle in the complex plane. Find λ.

Now let's have some fun.

Definition 2.3. A *cline* is a circle or a line.

This definition kind of makes sense if you view a line as a circle with infinite radius and center at infinity.

Definition 2.4. A function f is called a *Möbius transformation* if it is of the form

$$f(z) = \frac{az + b}{cz + d},$$

and f is nonconstant (or $ad - bc \neq 0$).

Note that any Möbius transformation is bijective. Now what do these two above definitions have anything to do with each other? Well, we have the following nuclear weapon.

Theorem 2.5. Möbius transformations take clines to clines. In other words, if f is a Möbius transformation and C is a cline, then the image of C under f is a cline.

In algebraic geometry, the Möbius transformations are "morphisms of $\mathbb{P}^1$." Don't worry, I don't know what that means either. We have another fun fact.

Theorem 2.6. Given complex numbers A, B, C, X, Y, and Z such that A, B, and C are pairwise distinct and X, Y, and Z are pairwise distinct, there's exactly one Möbius transformation sending A to X, B to Y, and C to Z.

This is rarely applicable, but when it is, it's so, so powerful.

Example 2.7 (Prize 2023/13)**.** Let

$$f(t) = \frac{(10 + 9i)t - 10 + 9i}{t + i},$$

where $i = \sqrt{-1}$. Let $P = f(0)$, $Q = f(2023)$, and $R = f(1)$. Determine $\sin^2(m\angle PQR)$.

Example 2.8 (HMMT Team 2022/6)**.** Let $P(x) = x^4 + ax^3 + bx^2 + x$ be a polynomial with four distinct roots that lie on a circle in the complex plane. Prove that $ab \neq 9$.

Out of 99 teams and around 600 competitors, only *one* team solved the above problem in-contest, making it the hardest problem on the team round that year, despite its placement.

3 Problems

Problem 3.1 (Prize 2025/8)**.** Compute

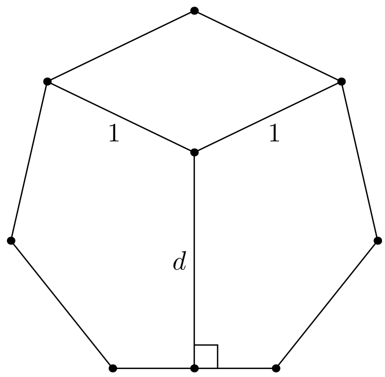
$$\sum_{n=0}^{\infty} \frac{\cos(\frac{\pi n}{6})}{\sqrt{3}^n}.$$

Problem 3.2 (Prize 2025/13)**.** Compute A if $-90 \le A \le 90$ and

$$\sin(A^\circ) = \frac{\frac{1}{2} + \sum_{k=0}^{44} \sin(2k^\circ)}{\sum_{k=0}^{44} \sin((2k+1)^\circ)}.$$

Problem 3.3 (CMIMC 2019/8)**.** The roots of the polynomial $P(z) = z^{2019} - 1$ can be written in the form $z_k = x_k + iy_k$ for $1 \le k \le 2019$. Let Q denote the monic polynomial with roots equal to $2x_k + iy_k$ for $1 \le k \le 2019$. Compute $Q(-2)$.

Problem 3.4 (Prize 2019/16)**.** The figure shows a regular heptagon with sides of length 1.



Determine the indicated length d. Express your answer in simplified radical form.